

YOUSEF ELBEYALY

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EDUCATION

Faculty of Computer Science
Bachelor of Computer Science

2025 – 2029 (*Expected*)
Egypt

- **Technical Foundation:** Linear Algebra, Multivariate Calculus, Probability & Statistics, Numerical Analysis.
- **Core CS:** Deep Learning (PyTorch), Machine Learning, Data Structures, Stochastic Processes.

TECHNICAL SKILLS

<i>Languages</i>	Python (Advanced), C#, SQL, HTML/CSS
<i>Frameworks</i>	PyTorch, TensorFlow, Keras, OpenCV, MediaPipe, Scikit-learn, CatBoost
<i>Tools</i>	Docker, NVIDIA TensorRT (Learning), Git/GitHub, Hugging Face, Flask, FastAPI, GCP
<i>Expertise</i>	Neural Network Architecture, Model Optimization, Object Detection & Segmentation

EXPERIENCE

Freelance Machine Learning Engineer
Self-Employed (Remote)

Jan 2026 – Present

- Designing end-to-end Computer Vision and Deep Learning solutions, specializing in production-ready APIs.
- Researching advanced architectures like **Flow Matching** and **Diffusion Models** to optimize generative workflows.
- Implementing model optimization techniques for high-performance inference.

SELECTED AI PROJECTS

Chest Cancer Detection (Deep Learning/CV)

PyTorch, CNN, VGG16

- Built a high-precision classification system for medical X-ray images using Transfer Learning.
- Applied custom Data Augmentation and Fine-tuning to handle imbalanced medical datasets.
- Optimized pre-processing pipelines with OpenCV for consistent model input.

AI Virtual Mouse & Trainer (Computer Vision)

OpenCV, MediaPipe, NumPy

- Engineered a real-time Human-Computer Interaction (HCI) system using hand landmark detection.
- Developed custom logic for real-time form-analysis and movement tracking for fitness monitoring.
- Optimized video stream processing to ensure low-latency performance on CPU/GPU.

AI Mizan (Forecasting)

CatBoost, Flask, Hugging Face

- Developed a predictive model for commodity prices with an **R2 score of 90.8%**.
- Deployed the full-stack pipeline using Flask and Hugging Face Spaces for public accessibility.

THEORETICAL FOUNDATIONS

- Deep understanding of **Numerical Optimization** and its application in Neural Network training.
- Proficient in **Backpropagation** mechanics and custom training loop implementations in PyTorch.